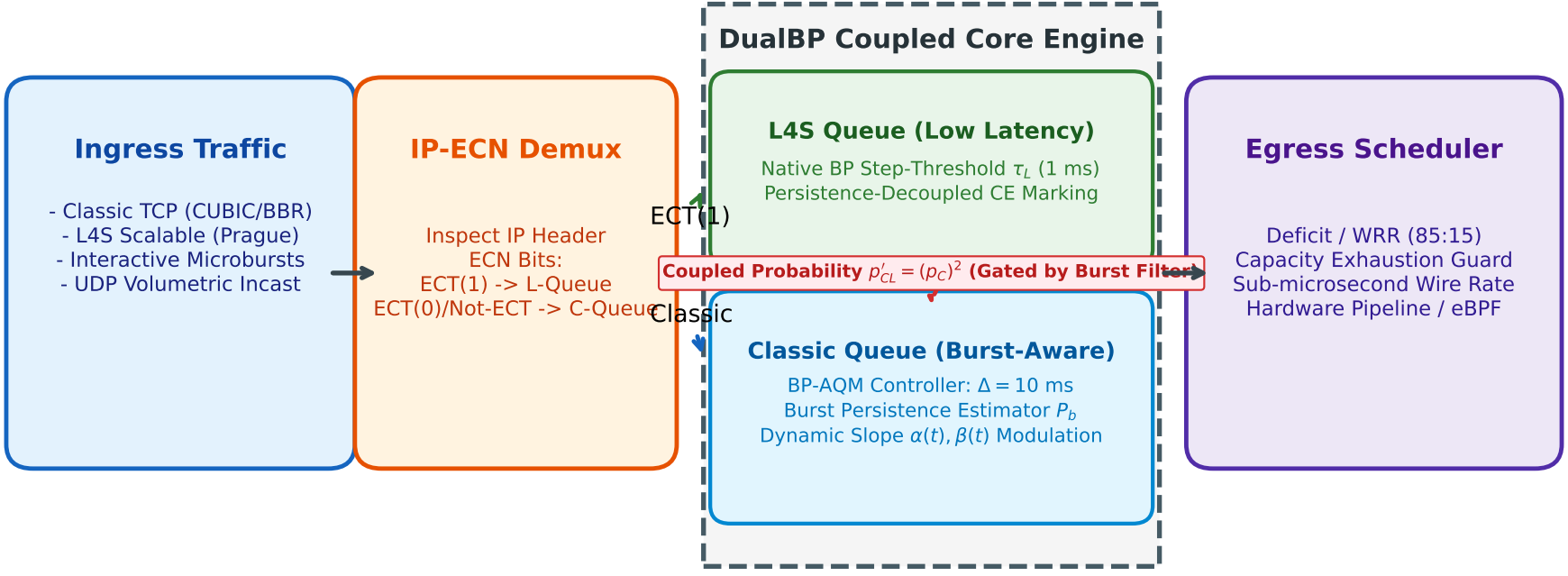
Ingress Traffic
- Classic TCP (CUBIC/BBR)
- L4S Scalable (Prague)
- Interactive Microbursts
- UDP Volumetric Incast
IP-ECN Demux
Inspect IP Header
ECN Bits:
ECT(1) -> L-Queue
ECT(0)/Not-Ect -> C-Queue
ECT(1)
Classic
DualBP Coupled Core Engine
L4S Queue (Low Latency)
Native BP Step-Threshold τ_L (1 ms)
Persistence-Decoupled CE Marking
Coupled Probability $p'_{CL} = (p_C)^2$ (Gated by Burst Filter)
Classic Queue (Burst-Aware)
BP-AQM Controller: $\Delta = 10$ ms
Burst Persistence Estimator P_b
Dynamic Slope $\alpha(t), \beta(t)$ Modulation
Egress Scheduler
Deficit / WRR (85:15)
Capacity Exhaustion Guard
Sub-microsecond Wire Rate
Hardware Pipeline / eBPF